Trodimenzijske kontinuirane slučajne varijable – zavisne

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1 Funkcija gustoće vjerojatnosti

$$f(x,y,z) = \frac{2}{3}(x+y+z), \qquad 0 \le x,y,z \le 1 \qquad (0 \text{ inače})$$

Provjera normiranosti:

$$\int_0^1 \int_0^1 \int_0^1 \frac{2}{3}(x+y+z)\,dx\,dy\,dz = \frac{2}{3}\left(\frac{1}{2}+\frac{1}{2}+\frac{1}{2}\right) = 1 \quad \checkmark$$

2 Marginalne funkcije gustoće

$$f(x) = \int_0^1 \int_0^1 \frac{2}{3}(x+y+z)\,dy\,dz = \frac{2}{3}x + \frac{2}{3} \tag{1}$$

$$f(y) = \int_0^1 \int_0^1 \frac{2}{3}(x+y+z)\,dx\,dz = \frac{2}{3}y + \frac{2}{3} \tag{2}$$

$$f(z) = \int_0^1 \int_0^1 \frac{2}{3}(x+y+z)\,dx\,dy = \frac{2}{3}z + \frac{2}{3} \tag{3}$$

$$f(x)f(y)f(z) \neq f(x,y,z) \;\longrightarrow\; \text{zavisne varijable}$$

Dvodimenzijske marginalne gustoće:

$$f(x,y) = \int_0^1 \frac{2}{3}(x+y+z)\,dz = \frac{2}{3}x + \frac{2}{3}y + \frac{1}{3} \neq f(x)f(y) \tag{4}$$

$$f(x,z) = \int_0^1 \frac{2}{3}(x+y+z)\,dy = \frac{2}{3}x + \frac{2}{3}z + \frac{1}{3} \neq f(x)f(z) \tag{5}$$

$$f(y,z) = \int_0^1 \frac{2}{3}(x+y+z)\,dx = \frac{2}{3}y + \frac{2}{3}z + \frac{1}{3} \neq f(y)f(z) \tag{6}$$

3 Uvjetne funkcije gustoće

Opća formula: $f(x \mid y,z) = \dfrac{f(x,y,z)}{f(y,z)}$. Budući da su varijable *zavisne*, nazivnik je dvodimenzijska marginalna gustoća $f(y,z)$ i **ne smije** se zamijeniti produktom $f(y)f(z)$, jer je $f(y,z) \neq f(y)f(z)$.

$$f(x \mid y,z) = \frac{f(x,y,z)}{f(y,z)} = \frac{\frac{2}{3}(x+y+z)}{\frac{2}{3}y+\frac{2}{3}z+\frac{1}{3}} = \frac{x+y+z}{y+z+\frac{1}{2}} = \frac{2(x+y+z)}{2y+2z+1} \tag{7}$$

$$\neq f(x) \;\longrightarrow\; \text{zavisne varijable}$$

Kontrola normiranosti (po x): $\int_0^1 \frac{x+y+z}{y+z+\frac{1}{2}}\,dx = \frac{\frac{1}{2}+y+z}{y+z+\frac{1}{2}} = 1.$ ✓

$$f(y \mid x,z) = \frac{f(x,y,z)}{f(x,z)} = \frac{x+y+z}{x+z+\frac{1}{2}} = \frac{2(x+y+z)}{2x+2z+1} \neq f(y) \tag{8}$$

$$f(z \mid x,y) = \frac{f(x,y,z)}{f(x,y)} = \frac{x+y+z}{x+y+\frac{1}{2}} = \frac{2(x+y+z)}{2x+2y+1} \neq f(z) \tag{9}$$

$$f(x,y \mid z) = \frac{f(x,y,z)}{f(z)} = \frac{\frac{2}{3}(x+y+z)}{\frac{2}{3}z+\frac{2}{3}} = \frac{x+y+z}{z+1} \neq f(x)f(y) \tag{10}$$

$$f(x,z \mid y) = \frac{f(x,y,z)}{f(y)} = \frac{x+y+z}{y+1} \neq f(x)f(z) \tag{11}$$

$$f(y,z \mid x) = \frac{f(x,y,z)}{f(x)} = \frac{x+y+z}{x+1} \neq f(y)f(z) \tag{12}$$

4 Matematička očekivanja i varijance

$$E(X) = \int_a^b x f(x)\,dx = \int_0^1 x\left(\frac{2}{3}x+\frac{2}{3}\right)dx = \frac{5}{9} \tag{13}$$

ili, budući da je $f(x) = \int_0^1\int_0^1 \frac{2}{3}(x+y+z)\,dy\,dz$,

$$E(X) = \int_0^1\int_0^1\int_0^1 x\cdot\frac{2}{3}(x+y+z)\,dx\,dy\,dz = \frac{5}{9} \tag{14}$$

$$E(Y) = \int_0^1 y\left(\frac{2}{3}y+\frac{2}{3}\right)dy = \frac{5}{9} \tag{15}$$

$$E(Z) = \int_0^1 z\left(\frac{2}{3}z+\frac{2}{3}\right)dz = \frac{5}{9} \tag{16}$$

$$V(X) = \int_a^b x^2 f(x)\,dx - E(X)^2 = \int_0^1 x^2\left(\frac{2}{3}x+\frac{2}{3}\right)dx - \left(\frac{5}{9}\right)^2 = \frac{13}{162} \tag{17}$$

$$V(Y) = \int_0^1 y^2\left(\frac{2}{3}y+\frac{2}{3}\right)dy - \left(\frac{5}{9}\right)^2 = \frac{13}{162} \tag{18}$$

$$V(Z) = \int_0^1 z^2\left(\frac{2}{3}z+\frac{2}{3}\right)dz - \left(\frac{5}{9}\right)^2 = \frac{13}{162} \tag{19}$$

5 Uvjetno očekivanje, kovarijance i koeficijenti korelacije

5.1 Uvjetno očekivanje

$$E(X \mid y,z) = \int_a^b x f(x \mid y,z)\,dx = \int_0^1 x\,\frac{x+y+z}{y+z+\frac{1}{2}}\,dx \tag{20}$$

$$= \frac{1}{y+z+\frac{1}{2}}\int_0^1 \left(x^2+x(y+z)\right)dx = \frac{\frac{1}{3}+\frac{y+z}{2}}{y+z+\frac{1}{2}} \tag{21}$$

$$E(X \mid y,z) = \frac{3y+3z+2}{6y+6z+3} \longrightarrow \text{zavisne varijable, regresija nije linearna} \tag{22}$$

Kontrola: za $y = z = 0$ je $f(x \mid 0,0) = \dfrac{x}{1/2} = 2x$, pa je $E(X \mid 0,0) = \dfrac{2}{3}$, što formula i daje: $\dfrac{3\cdot 0 + 3\cdot 0 + 2}{6\cdot 0 + 6\cdot 0 + 3} = \dfrac{2}{3}$. ✓

$$E(Y \mid x,z) = \int_0^1 y\,\frac{x+y+z}{x+z+\frac{1}{2}}\,dy = \frac{3x+3z+2}{6x+6z+3} \tag{23}$$

$$E(Z \mid x,y) = \int_0^1 z\,\frac{x+y+z}{x+y+\frac{1}{2}}\,dz = \frac{3x+3y+2}{6x+6y+3} \tag{24}$$

5.2 Kovarijance

$$\operatorname{Cov}(X,Y) = \int_a^b \int_c^d xy\,f(x,y)\,dx\,dy - E(X)E(Y) \tag{25}$$

$$= \int_0^1 \int_0^1 xy\left(\frac{2}{3}x + \frac{2}{3}y + \frac{1}{3}\right)dx\,dy - \left(\frac{5}{9}\right)^2 \tag{26}$$

$$= \frac{11}{36} - \left(\frac{5}{9}\right)^2 = \frac{99}{324} - \frac{100}{324} \tag{27}$$

$$\operatorname{Cov}(X,Y) = -\frac{1}{324} \tag{28}$$

ili

$$\operatorname{Cov}(X,Y) = \int_a^b \int_c^d \int_e^f \big(x - E(X)\big)\big(y - E(Y)\big) f(x,y,z)\,dx\,dy\,dz \tag{29}$$

$$= \int_0^1 \int_0^1 \int_0^1 \left(x - \frac{5}{9}\right)\left(y - \frac{5}{9}\right)\frac{2}{3}(x+y+z)\,dx\,dy\,dz = -\frac{1}{324} \tag{30}$$

ili

$$\operatorname{Cov}(X,Y) = E(XY) - E(X)E(Y) \tag{31}$$

$$E(XY) = \int_0^1 \int_0^1 \int_0^1 xy\,\frac{2}{3}(x+y+z)\,dx\,dy\,dz = \frac{11}{36} \tag{32}$$

$$\operatorname{Cov}(X,Y) = \frac{11}{36} - \frac{5}{9}\cdot\frac{5}{9} = -\frac{1}{324} \tag{33}$$

$$\operatorname{Cov}(X,Z) = -\frac{1}{324}, \qquad \operatorname{Cov}(Y,Z) = -\frac{1}{324}$$

5.3 Koeficijenti korelacije

$$\rho(X,Y) = \frac{\operatorname{Cov}(X,Y)}{\sigma(X)\,\sigma(Y)} = \frac{-\frac{1}{324}}{\sqrt{\frac{13}{162}}\cdot\sqrt{\frac{13}{162}}} = \frac{-\frac{1}{324}}{\frac{13}{162}} = -\frac{1}{26} \tag{34}$$

$$\rho(X,Z) = \frac{\operatorname{Cov}(X,Z)}{\sigma(X)\,\sigma(Z)} = -\frac{1}{26}, \qquad \rho(Y,Z) = \frac{\operatorname{Cov}(Y,Z)}{\sigma(Y)\,\sigma(Z)} = -\frac{1}{26}$$

Napomena: Koeficijent korelacije ρ uvijek je dobro definiran i mjeri *linearnu* povezanost varijabli. Budući da funkcije regresije ovdje nisu linearne, ρ opisuje samo linearnu komponentu zavisnosti, pa ga treba interpretirati s oprezom.